

Artificial Intelligence

Course Code: CS-202

Topic: Intelligent Agents and their types, Rationality

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Today's Goal

- Intelligent Agents
- Characteristics of Intelligent Agents
- Agents Interaction Cycle
- Why study intelligent agents?
- Rationality
- PEAS



This week lecture will cover the following CLO:

- **CLO1:** Discuss the key concepts in the field of artificial intelligence



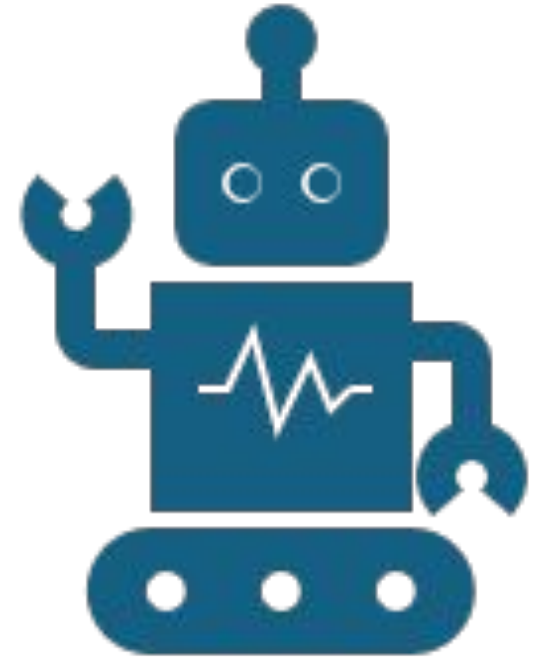
Intelligent Agent

Intelligent Agents

- What is an agent?
 - An **agent** is anything that
 - **perceiving** its environment through **sensors**
 - **acting** upon that environment through **actuators**
 - Example:
 - Human is an agent
 - A robot is also an agent with cameras and motors
 - A thermostat detecting room temperature

Human agent: eyes, ears, and other organs for sensors; - hands, legs, mouth, and other body parts for actuators

Robotic agent: cameras and infrared range finders for sensors -various motors for actuators



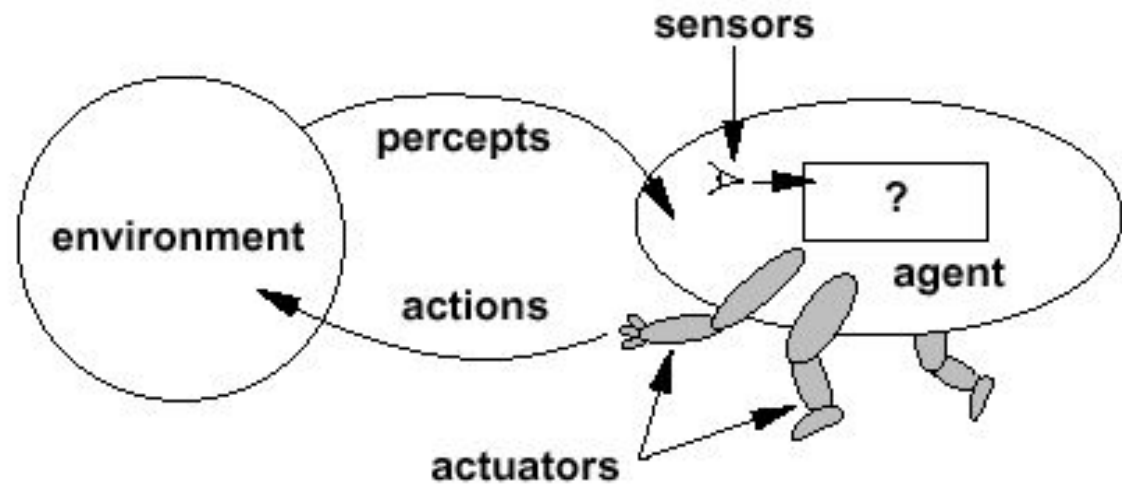
Intelligent Agents

- The agent function maps from percept histories to actions:

$$[f: P^* \rightarrow A]$$

- The agent program runs on the physical architecture to produce f

agent = architecture +
program



Examples of Intelligent Agents



Self-Driving Cars

- Perceive road conditions and make driving decisions.

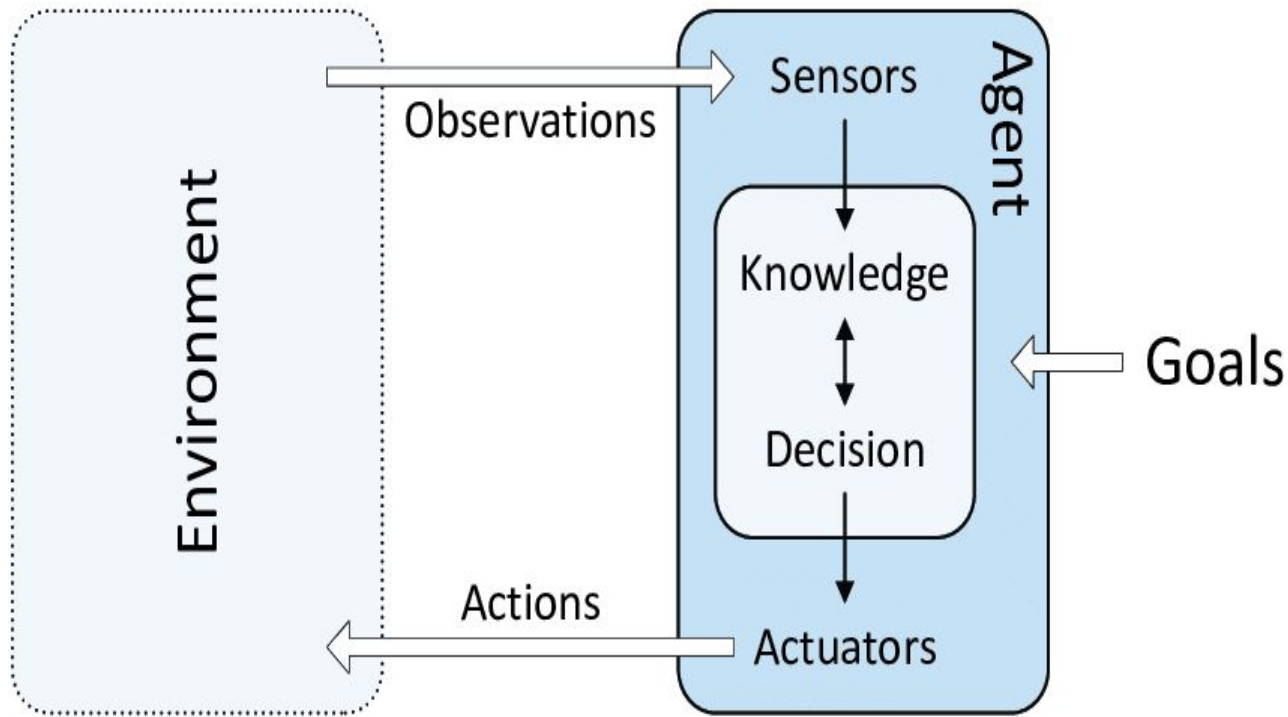
Virtual Personal Assistants (e.g., Siri, Alexa)

- Respond to voice commands and perform tasks.

Robots in Manufacturing

- Automate tasks by perceiving object positions and adjusting actions.

Agent's Interaction Cycle



Perception

- Collects information about the environment.

Reasoning/Decision Making

- Uses the information to make decisions.

Action

- Performs actions to affect the environment.

Why Study Intelligent Agents?

- Understanding intelligent agents helps develop better AI systems that can solve complex, real-world problems.
- Foundation for AI applications across various fields like healthcare, finance, gaming, and robotics.



Characteristics of Intelligent Agents

Autonomy:

- Capable of making decisions without human intervention.

Reactivity:

- Responds to changes in the environment.

Proactivity:

- Takes initiative to achieve its goals.

Social Ability:

- Can interact with other agents or humans to fulfil its objectives.

The Concept of Rationality

Rationality

- Rationality means making the best possible decision based on the information available. A rational agent chooses actions that lead to the best expected outcome.
- **Factors influencing rationality:**
 - Performance Measure – How success is defined for the agent.
 - Percept Sequence – What the agent has sensed or observed over time.
 - Knowledge of the Environment – What the agent knows about the world.
 - Available Actions – The choices the agent can make.

Goals/Preferences

Prior Knowledge

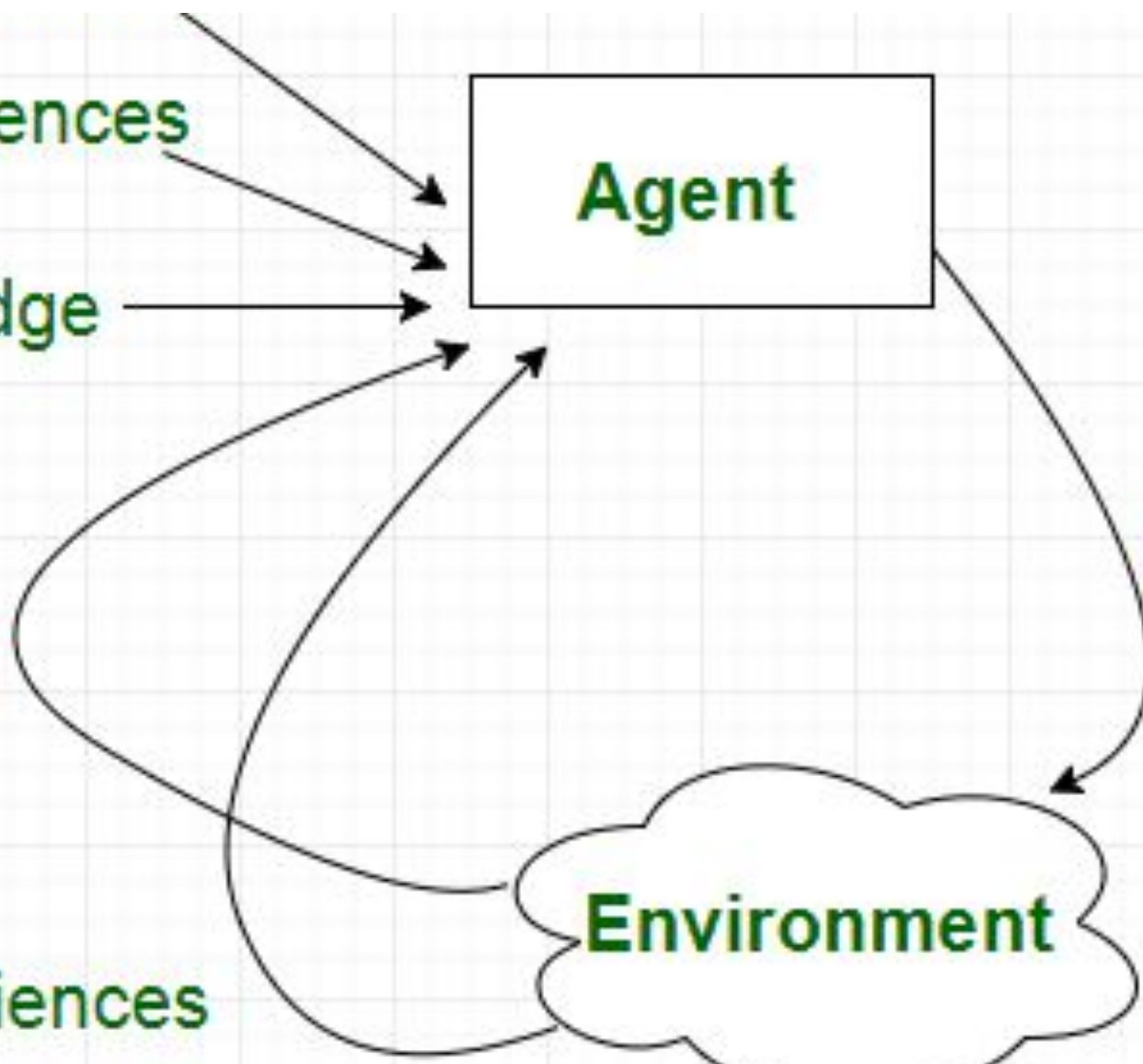
Observations

Past Experiences

Agent

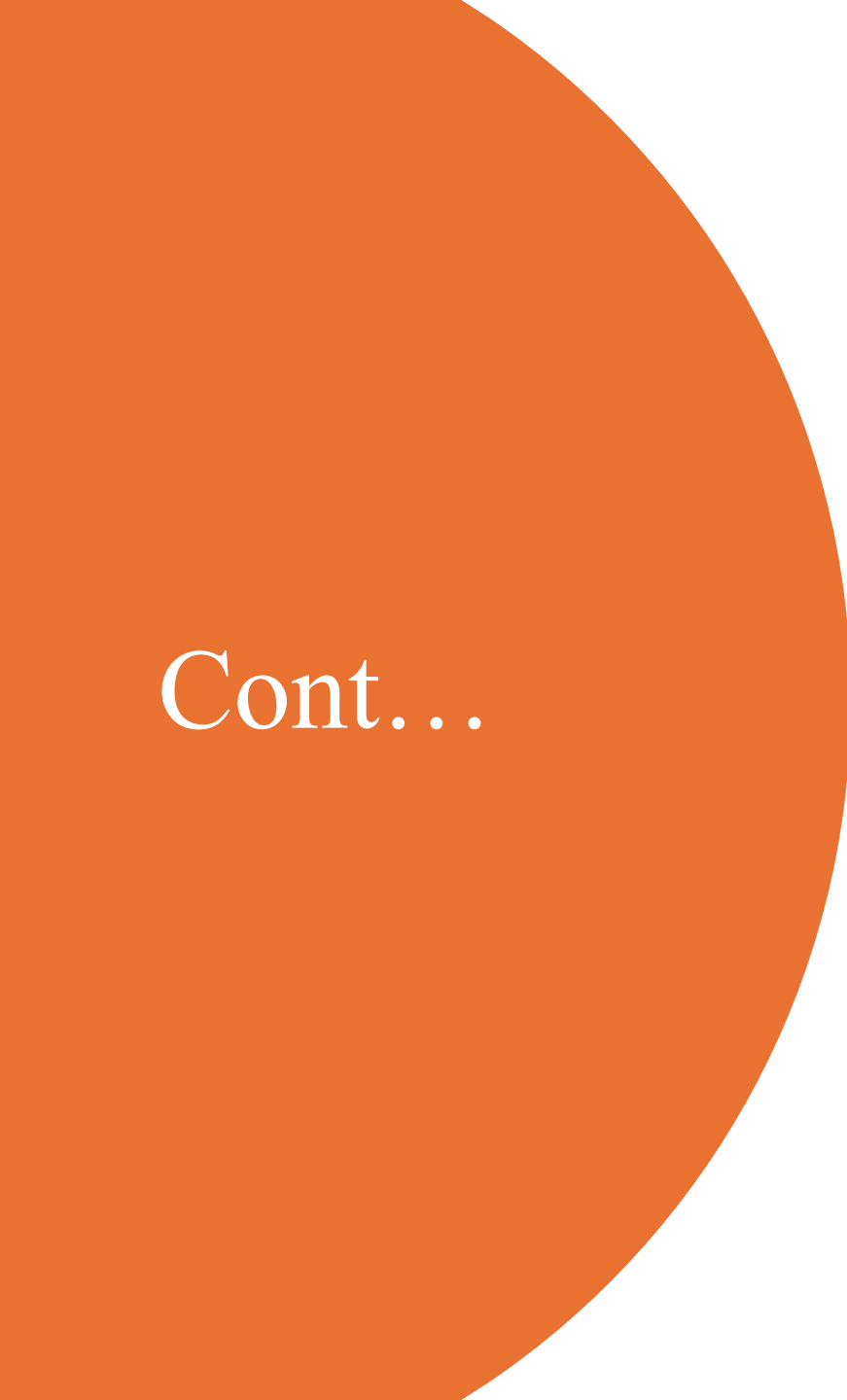
Environment

Actions



PEAS

- PEAS is a framework used to define the structure of an intelligent agent by identifying four key components:
 - **Performance Measure** – The criteria that define the success of the agent's actions (e.g., accuracy, speed, efficiency).
 - **Environment** – The surroundings in which the agent operates (e.g., a chessboard, a self-driving car's road).
 - **Actuators** – The parts that allow the agent to take action (e.g., robot arms, car wheels).
 - **Sensors** – The parts that gather information from the environment (e.g., cameras, microphones).



Cont...

- The **performance measure** determines what counts as a good decision, while the **environment, actuators, and sensors** influence how well the agent can perceive and act.
 - A rational agent uses these components to make the best possible choices.
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Nature of environments

- A sketch of automated taxi driver

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits, minimize impact on other road users	Roads, other traffic, police, pedestrians, customers, weather	Steering, accelerator, brake, signal, horn, display, speech	Cameras, radar, speedometer, GPS, engine sensors, accelerometer, microphones, touchscreen

Figure 2.4 PEAS description of the task environment for an automated taxi driver.

Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments	Touchscreen/voice entry of symptoms and findings
Satellite image analysis system	Correct categorization of objects, terrain	Orbiting satellite, downlink, weather	Display of scene categorization	High-resolution digital camera
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, tactile and joint angle sensors
Refinery controller	Purity, yield, safety	Refinery, raw materials, operators	Valves, pumps, heaters, stirrers, displays	Temperature, pressure, flow, chemical sensors
Interactive English tutor	Student's score on test	Set of students, testing agency	Display of exercises, feedback, speech	Keyboard entry, voice

Figure 2.5 Examples of agent types and their PEAS descriptions.

Any Question?

